Explainable Predictive Analytics for Employee Attrition Risk Assessment

An Explainable AI (XAI) & HR Decision-Support System Powered by ML and SHAP

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ABSTRACT

Employee attrition increases recruitment costs, disrupts project continuity and causes the loss of organizational knowledge. Traditional predictive systems can estimate who may leave, but their black-box outputs provide limited support for responsible human-resource decisions. This project presents an explainable predictive analytics system for employee attrition risk assessment that combines machine learning, SHAP-based interpretability and an interactive decision-support application. The system uses the IBM HR Analytics benchmark dataset containing 1,470 employee records and 35 attributes. Its preprocessing workflow applies one-hot encoding, standardization and an 80/20 stratified train-test split. Logistic Regression, Decision Tree, Random Forest and Gradient Boosting models are evaluated on 294 held-out samples. Random Forest is selected for the application because it provides stable generalization and direct compatibility with SHAP TreeExplainer, achieving 82.65% accuracy, 45.24% precision, 40.43% recall, a 42.70% F1-score and 78.55% ROC-AUC. SHAP explanations identify work-related drivers such as overtime, stock-option level, age, monthly income and manager tenure, while local attributions show how individual features increase or reduce a predicted risk score. A React interface and FastAPI service provide an executive risk dashboard, a filterable workforce directory, employee-level explanations, model benchmarking, a counterfactual what-if simulator and retention-action tracking backed by SQLite through SQLAlchemy. The system supports human review and proactive retention planning; it does not treat model attribution as causation and must not automate firing, promotion or disciplinary decisions.

Keywords: employee attrition; explainable AI; SHAP; random forest; predictive analytics; HR decision support; what-if simulation.